CAPITAL IN CODE / CIC-001

The Overnight Effect in SPY

A reproducible historical return decomposition of the close-to-next-open and open-to-close intervals.

RESEARCH QUESTION

From 1994-01-01 through 2025-12-31, did SPY historically earn more of its adjusted close-to-close return overnight than during regular market hours?

ABSTRACT

Using adjusted daily OHLC data for SPY, this study separates the previous trading close-to-next-open return from the open-to-close return. Across 8,053 aligned observations, overnight CAGR was 9.86% and regular-hours CAGR was 0.79%. Overnight had the highest Sharpe ratio at 0.929, using the configured 0.00% annual risk-free rate (default 0.00%). Paired tests and a bootstrap provide historical evidence of a positive average return difference under the study assumptions.

The pattern is not a trading recommendation. It varied across time, was sharply negative during the defined COVID shock window, and was substantially weakened by a simplified daily transaction-cost model. The results describe historical association in one adjusted-price sample - not causality, prediction, or executable profit.

8,053	1994-01-04	SPY
aligned return observations	through 2025-12-31	adjusted daily OHLC

Prepared by Dillon Nagar | Verified fixed-sample results

01 / DESIGN AND FINDINGS

A focused question, paired evidence, and explicit limits.

The study uses Yahoo Finance daily SPY Open and Close data through yfinance with *auto_adjust=True*. The data convention applies a consistent split- and dividend-adjustment factor to both fields; raw and adjusted values are never mixed. The verified run contained 8,054 clean price rows and 8,053 aligned return rows, with no invalid prices, duplicate dates, or removals beyond the first unaligned observation.

For each observed trading day t, overnight return equals current Open divided by the previous observed Close minus 1. Regular-hours return equals current Close divided by current Open minus 1. The pipeline verifies that the components reconstruct the adjusted close-to-close return on every row.

FULL-SAMPLE RETURN DECOMPOSITION

Metric	Overnight	Regular hours	Buy and hold
Daily standard deviation	0.6769%	0.9721%	1.1867%
Cumulative return	1917.13%	28.63%	2494.63%
Annualized return (CAGR)	9.86%	0.79%	10.73%
Annualized volatility	10.75%	15.43%	18.84%
Sharpe ratio (0.00% RF)	0.929	0.128	0.635
Maximum drawdown	-32.79%	-68.49%	-55.19%
Win rate (positive days)	56.29%	52.63%	54.28%
CAGR rank	2	3	1
Sharpe rank	1	3	2
Excess CAGR vs. buy and hold	-0.87 pp	-9.94 pp	+0.00 pp
High return / high volatility	No	No	Yes

The overnight component has the highest Sharpe ratio, while buy and hold has the highest CAGR. Ranks are descriptive and use only the return streams shown in this table; they are not strategy selection or evidence of executable results. The 1 bp-per-leg transaction-cost model remains a separate sensitivity.

02 / RISK-ADJUSTED COMPARISON

Return needs a volatility reference.

The chart compares each return stream's CAGR with annualized volatility, calculated as sample daily standard deviation times sqrt(252). The Sharpe ratio uses a 0.00% annual risk-free rate. The outlined point meets the descriptive high-return/high-volatility rule: upper half by CAGR and higher half by annualized volatility (Buy and hold).

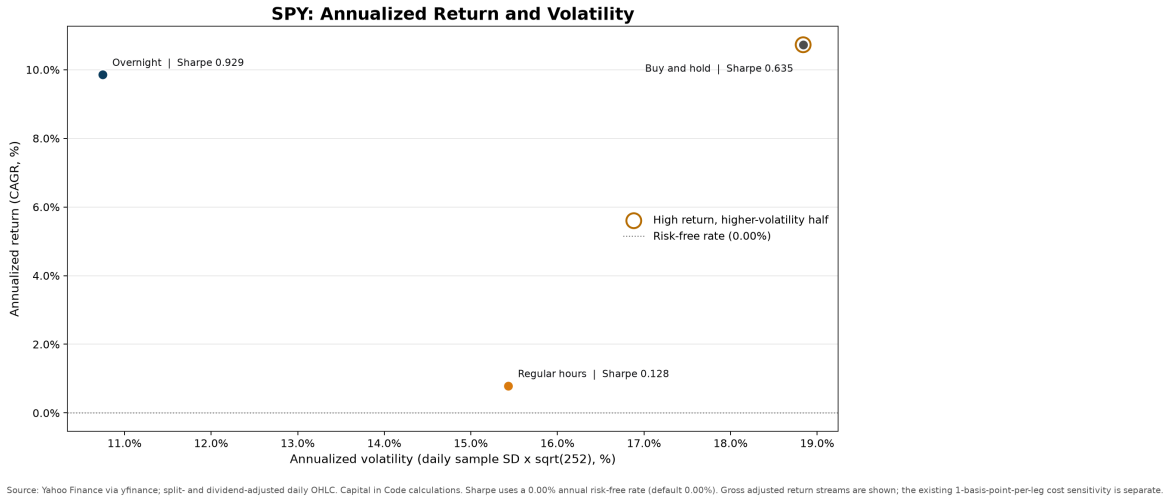


Figure 1. CAGR versus annualized volatility. The chart uses gross adjusted returns; transaction costs are not included.

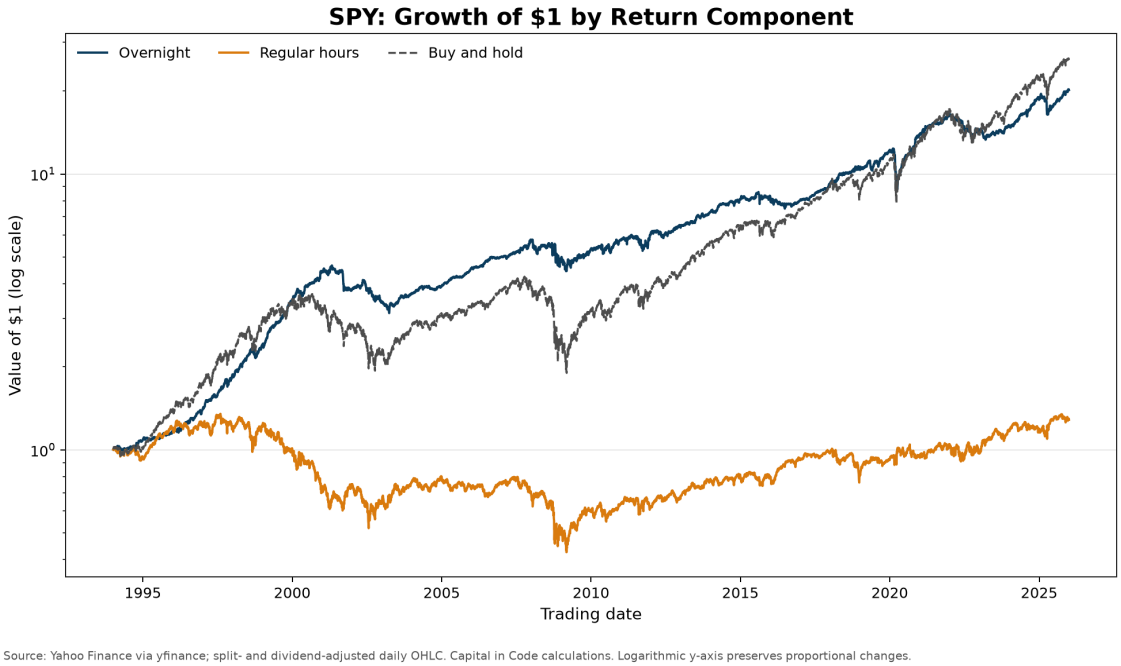


Figure 2. Compounded adjusted growth of $1 by component. The logarithmic axis preserves proportional changes.

03 / INFERENCE AND ROBUSTNESS

Statistical evidence is not the same as economic feasibility.

The primary comparison uses matched overnight and regular-hours returns from the same trading dates. A paired design reflects that the two components are parts of the same day. The paired t-test evaluates the mean difference; the Wilcoxon signed-rank test is a nonparametric paired check; and a paired bootstrap resamples same-date pairs 10,000 times with a fixed seed.

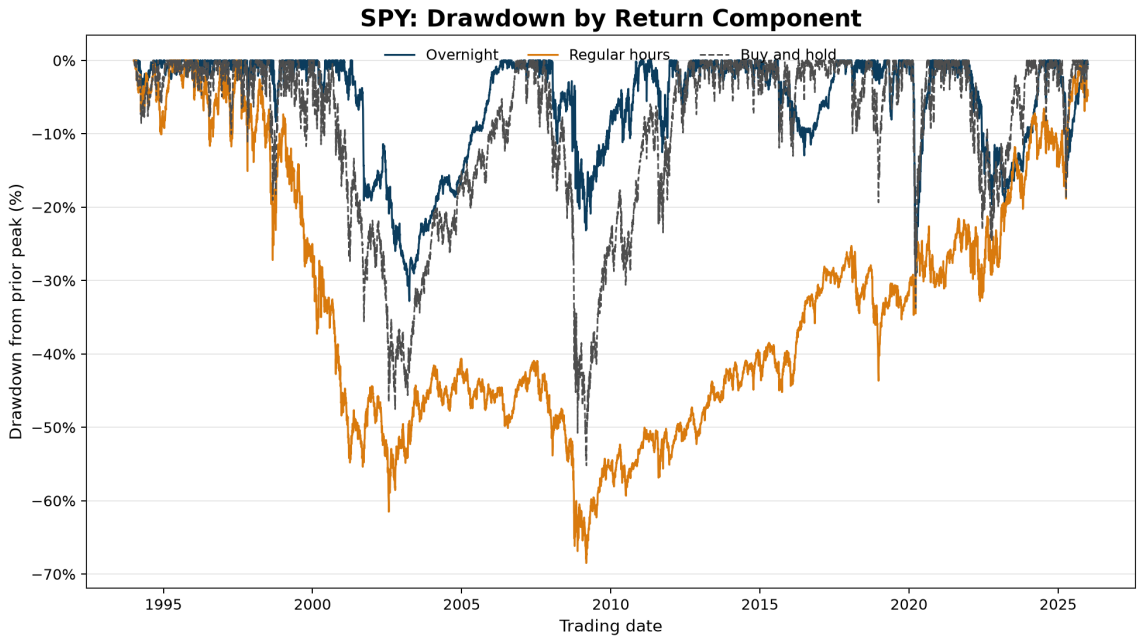
Measure	Verified result	Interpretation
Mean paired difference	3.18 bps/day	Overnight minus regular hours
Paired t-test	p = 0.0160	Evidence under its null model
Wilcoxon signed-rank	p = 0.000162	Nonparametric paired check
Paired bootstrap	0.55 to 5.74 bps/day	95% interval for mean difference

Statistical significance concerns compatibility with a test's null model. Economic significance additionally requires attention to magnitude, risk, costs, capacity, and investability.

ROBUSTNESS SNAPSHOT

Sensitivity check	Overnight	Regular hours	What it shows
Later start: 2000	6.96%	1.01%	Ordering remains, gap is smaller
Later start: 2010	8.62%	4.89%	Period dependence is material
Later start: 2020	8.71%	5.66%	Recent gap remains narrower
COVID shock	-28.12%	-7.34%	Overnight exposure was especially damaging

A paired tail trim retained 7,769 dates after removing 284 dates outside the components' respective 1st and 99th percentiles. The retained sample still favored overnight (3,746.17% compounded versus 11.76%), but this is a sensitivity check - not a claim that an investor could avoid adverse tails.



Source: Yahoo Finance via yfinance; split- and dividend-adjusted daily OHLC. Capital in Code calculations.

Figure 2. Drawdown from each component's prior peak. Component wealth paths are not simulated executable strategies.

04 / PRACTICAL INTERPRETATION

The evidence describes a pattern. It does not prove a strategy.

The transaction-cost exercise applies a deliberately simple 1-basis-point cost on entry and the same cost on exit for every observed trading day. It assumes fills at price multiplied by (1 + cost) on entry and price multiplied by (1 - cost) on exit. This mechanical sensitivity is useful for scale, but it is not an execution simulation.

Component	Gross annualized	Net annualized	Net cumulative
Overnight	9.86%	4.46%	302.96%
Regular hours	0.79%	-4.16%	-74.30%

Even under this simplified model, the overnight series remains positive, but it underperformed the 10.73% annualized historical close-to-close buy-and-hold result. The model omits changing spreads, slippage, market impact, partial or failed executions, financing, account constraints, capacity, and taxes.

LIMITATIONS

Data	Yahoo Finance may revise adjusted history; yfinance is an unofficial interface to that provider.
Pricing	Adjusted daily Open and Close are historical accounting inputs, not evidence of executable prices at a chosen size or time.
Statistics	Daily differences can have serial dependence, fat tails, and changing volatility not fully modeled by these procedures.
Scope	One U.S. ETF and one fixed sample do not establish cross-asset stability, causality, or future performance.

REPRODUCIBILITY

The full source code, methods, tests, and generated result artifacts are available in the public Capital in Code repository. The documented configuration can be reproduced from the CIC-001 project directory with `python -m src.main --ticker SPY --start-date 1994-01-01 --end-date 2025-12-31`. The fixed bootstrap seed is 42 with 10,000 resamples. A future refresh may differ if Yahoo Finance revises historical data or adjustment logic.

Conclusion: within this adjusted SPY sample, more historical close-to-close growth was associated with the close-to-next-open interval than with regular trading hours. That finding is historical evidence about a return decomposition, not a causal explanation, forecast, or investment recommendation.

Sources: Yahoo Finance daily SPY OHLC data retrieved via yfinance; CIC-001 generated result artifacts; CIC-001 research pipeline. Full formal report and source materials: github.com/Manateek1/Capital-in-Code